Exercise 2.4

1. 4
2. $8C_4 + 8C_3 = 9C_4 = 126$
3. 210 cards
4. 20
5. 25200
6. 671
7. $\frac{13!}{7!\,6!} \times 9! \times 9!$
8. 11
9. (i) 1365 (ii) 1001 (iii) 364
10. (i) 186 (ii) 186

Exercise 2.6

1. (i) $16a^4 - 96a^3b + 216a^2b^2 - 216ab^3 + 81b^4$

   (ii) $x^7 + \frac{7x^6}{y} + \frac{21x^5}{y^2} + \frac{35x^4}{y^3} + \frac{35x^3}{y^4} + \frac{21x^2}{y^5} + \frac{7x}{y^6} + \frac{1}{y^7}$

   (iii) $x^6 + 6x^3 + 15 + \frac{20}{x^3} + \frac{15}{x^6} + \frac{6}{x^9} + \frac{1}{x^{12}}$

2. (i) 10, 40,60,401 (ii) 995009990004999
3. 11440 $x^9 y^4$
4. (i) $11C_5 x$, $\frac{11C_5}{x}$ (ii) $\frac{8C_4(81)}{16}x^{12}$ (iii) $\frac{-10C_5(6)^5}{x^5}$
5. (i) $9C_6 \frac{2^6}{3^6}$ (ii) $-32(15C_5)$ (iii) 7920.

Exercise 2.7

1	2	3	4	5	6	7	8	9	10	11	12	13
(d)	(c)	(d)	(c)	(a)	(c)	(d)	(b)	(b)	(c)	(b)	(d)	(b)
14	15	16	17	18	19	20	21	22	23	24	25	
(a)	(c)	(a)	(c)	(a)	(c)	(a)	(b)	(c)	(d)	(b)	(a)	

Miscellaneous Problems

1. $\frac{3}{x-1} + \frac{2}{x+3}$
2. $\frac{3}{x-1} - \frac{2}{x-2}$
3. $\frac{1}{x+2} + \frac{5}{x-3} - \frac{3}{(x-3)^2}$
4. $\frac{3}{x-2} + \frac{2x-1}{x^2-x+1}$
5. (i) $\frac{7}{2}$ (ii) $\frac{7}{24}$ (iii) 12
6. (a) 720 (b) 7776 (c) 120 (d) 6
7. 74 ways
8. 85 ways
9. 210
10. 544

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